

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO5:

Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, wellstructured subproblems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Recommended Network Solution for a Rural Branch Office

The rural branch has constraints such as unavailability of high-speed wired internet, limited budget, need for uninterrupted operation, cloud applications, video conferencing, and secure communication.

1. Internet Connection:

Use a 4G/5G wireless broadband connection as the primary Internet connection. If available, a satellite Internet connection can be used in areas with poor cellular coverage.

2. Backup Connectivity:

Provide a second connection, such as another mobile network or satellite link, as a backup. Automatic failover ensures business operations continue if the primary connection fails.

3. Networking Devices:

- 4G/5G router or satellite modem
- Gigabit Ethernet switch
- Wi-Fi access points
- Enterprise router with dual-WAN support

4. Bandwidth Management:

Use Quality of Service (QoS) to prioritize video conferencing, cloud applications, and important business traffic. This prevents non-critical activities from consuming all available bandwidth.

5. Security:

Deploy a firewall at the branch router to block unauthorized traffic. Use VPN encryption for secure communication between the branch and headquarters. Strong passwords and secure Wi-Fi encryption should also be used.

6. Reliability and Cost:

A wireless primary connection with a low-cost backup provides a good balance between cost and reliability. Network monitoring can detect failures and automatically switch to the backup connection.

7. Future Scalability:

Choose equipment that supports higher bandwidth, additional users, extra access points, and future 5G/fiber availability. This allows the network to grow without completely replacing the existing infrastructure.

2. Wireless Network Design for a University

The university needs campus-wide wireless access for more than 3,000 students in classrooms, laboratories, libraries, and hostels. The major constraints are a large number of simultaneous users, limited access points, interference, security, and budget.

1. Access Point Placement:

Place wireless access points (APs) at strategic locations such as classrooms, libraries, laboratories, corridors, and hostels. APs should be positioned to provide maximum coverage with minimum overlap.

2. Frequency Band Selection:

Use 5 GHz wherever possible because it provides higher performance and has less interference than 2.4 GHz. Use 2.4 GHz for devices that require greater coverage.

3. Channel Planning:

Configure nearby APs to use different non-overlapping channels. This reduces co-channel and adjacent-channel interference.

4. High-Density Support:

Use enterprise-grade APs capable of handling many simultaneous users. Enable technologies such as band steering and load balancing to distribute users among available APs.

5. Authentication and Security:

Use WPA2/WPA3-Enterprise with username/password authentication. Integration with the university's authentication server can provide secure individual accounts.

6. Performance and Cost:

Since only a limited number of APs can be deployed, APs should be installed in high-

user-density areas first. Proper placement provides better coverage without unnecessarily increasing the number of devices.

7. Scalability and Management:

Use centralized wireless management to monitor APs, users, bandwidth, and interference. Additional APs can be added later as the number of students increases.

3. Secure Network Architecture for a Financial Institution :

The financial institution must protect sensitive customer information and online banking services while maintaining network performance, controlling costs, and meeting strict security requirements.

1. Network Segmentation:

Divide the network into separate VLANs/subnets for departments such as employees, servers, guest users, and critical banking systems. This prevents unauthorized access between different network sections.

2. Firewall Deployment:

Deploy a next-generation firewall (NGFW) between the internal network and the Internet. It should inspect traffic and block unauthorized connections. Additional firewall rules can protect sensitive server networks.

3. Authentication:

Use strong authentication such as multi-factor authentication (MFA) for employees accessing sensitive systems. Role-based access should ensure that users can access only the information required for their jobs.

4. Intrusion Detection:

Deploy an IDS/IPS to continuously monitor network traffic for suspicious activities, attacks, and unauthorized access attempts. Alerts can be generated when threats are detected.

5. Secure Communication:

Use VPN and encryption for remote employees and communication between important network locations. Sensitive customer data should be protected during transmission.

6. Performance and Cost:

Use VLANs and firewall rules efficiently rather than deploying expensive security devices everywhere. Centralized monitoring reduces administrative effort and helps identify problems quickly.

7. Compliance and Monitoring:

Maintain security logs and regularly monitor network activity. Access controls, encryption, authentication, and audit logs help the institution meet strict security and compliance requirements.